KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 4 REPORT – THE EDA DEEP DIVE

Univariate and Bivariate Analysis: Visualizing Distributions

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

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1. Introduction

This report presents the Week 4 work of the project "Youth Smoking and Drug Abuse Analysis". Week 4 is the "EDA Deep Dive", in which the cleaned dataset from the Week 3 Cleaning Sprint is explored in detail. The work covers univariate analysis (studying one variable at a time), bivariate analysis (studying the relationship between two variables) and the visualization of distributions using count plots, histograms, grouped count plots, boxplots, scatter plots and a correlation heatmap.

The analysis is carried out on the cleaned dataset (df_clean) of 1,885 records, which has no missing values and no duplicate rows. As decided earlier, outliers were reported and retained, so all genuine observations are part of the exploration.

2. Objective

To explore the distributions of individual variables and the relationships between pairs of variables in the cleaned dataset, using statistical summaries and suitable visualizations, so that patterns in the demographic, personality and drug-consumption variables can be understood before feature engineering.

3. Activities Performed

- Created a separate copy of the cleaned dataset (df_eda) for exploration so that df_clean remains unchanged.
- Converted the encoded Gender, Age and Education codes into readable labels for visualization.
- Performed univariate analysis of Gender, Age Group and Education Level using count plots.
- Studied the distribution of the seven personality scores using histograms.
- Studied the distribution of nine drug-consumption variables using count plots.
- Performed bivariate analysis of Age Group vs Gender, and of Gender, Age Group and Education Level vs Alcohol consumption.
- Compared Nscore across alcohol and cannabis consumption levels using boxplots.
- Examined relationships between personality scores using scatter plots and a correlation heatmap.

4. Approach and Data Preparation

4.1 Choosing the Right Plot

The plot type was selected according to the data types of the variables involved. Categorical variables were summarised with counts, numerical variables with histograms, and pairs of variables with the plot that suits their combination of types.

Analysis	Variable Types	Plot Used
Univariate	Categorical	Count plot
Univariate	Numerical	Histogram
Bivariate	Categorical vs Categorical	Grouped count plot
Bivariate	Numerical vs Categorical	Boxplot
Bivariate	Numerical vs Numerical	Scatter plot
Bivariate	All numerical pairs	Correlation heatmap

Table 1: Plot types selected for each kind of analysis

4.2 Converting Encoded Values into Labels

Gender, Age and Education are stored as numerical codes in the dataset, which are difficult to read on a chart. New label columns (Gender_Label, Age_Group and Education_Level) were created in df_eda for visualization only. The original encoded columns were retained, so the underlying data was not modified. In the education labels, the code equal to the median-imputed value was kept as a separate "Median Imputed" category so that imputed records are visible in the plot.

Variable	Encoded Value	Label
Gender	0.48246	Female
Gender	-0.48246	Male
Age	-0.95197	18-24
Age	-0.07854	25-34
Age	0.49788	35-44
Age	1.09449	45-54
Age	1.82213	55-64
Age	2.59171	65+

Table 2: Encoded values and their labels for Gender and Age

5. Univariate Analysis

5.1 Gender and Age Group

Gender has two categories. The dataset contains 1,006 female (53.4%) and 879 male (46.6%) participants, so the two groups are reasonably balanced with a slightly higher share of females. The Age Group plot shows that the participants are concentrated in the younger age groups: the 18–24 group is the largest (605 records, 32.1%), followed by 25–34 (565 records, 30.0%). Together these two groups form about 62% of the data. The count falls steadily with age, and the 65+ group has only 16 records (0.8%).

Variable	Category	Count	Percentage
Gender	Female	1,006	53.4%
Gender	Male	879	46.6%
Age Group	18-24	605	32.1%
Age Group	25-34	565	30.0%
Age Group	35-44	335	17.8%
Age Group	45-54	275	14.6%
Age Group	55-64	89	4.7%
Age Group	65+	16	0.8%

Table 3: Distribution of Gender and Age Group

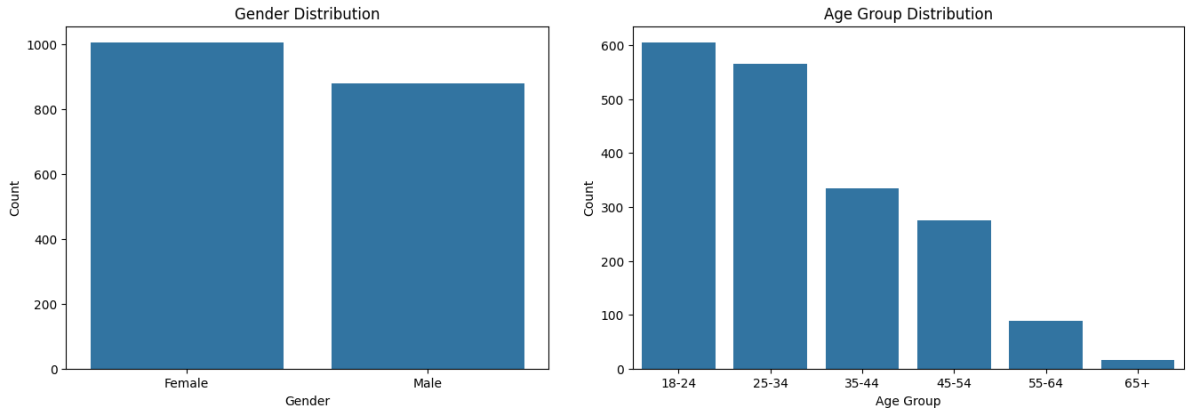


Figure 1: Count plots of Gender (left) and Age Group (right)

Because the 18–24 group is the largest age group in the data, the dataset is well suited to the youth-focused analysis planned for this project. The very small 65+ group means that any pattern seen for that group should be interpreted with caution.

5.2 Education Level

Education Level is an ordinal categorical variable. The count plot shows that the middle education categories have the highest counts, while the categories for leaving school early (before 16, at 16, at 17 and at 18) contain far fewer participants. The “Median Imputed” bar represents the records that carry the median-imputed education value, which includes the 113 records filled during Week 3, so it is larger than a typical single category.

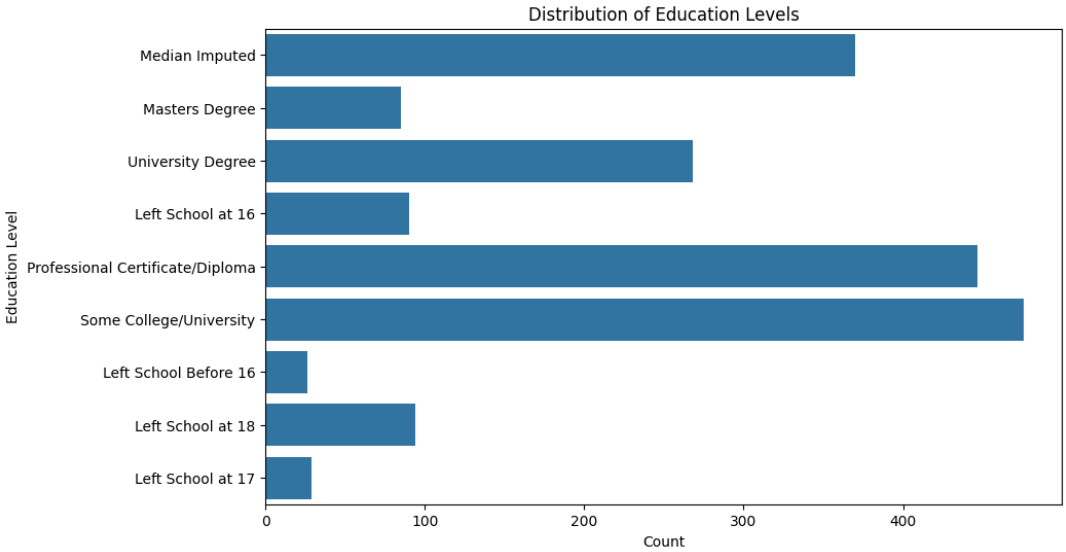


Figure 2: Distribution of Education Levels

5.3 Personality Score Distributions

Histograms were drawn for the seven personality scores: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. Nscore, Escore, Oscore, Ascore and Cscore are approximately bell-shaped and centred near zero with a spread of roughly −3 to +3, which is consistent with scores that were standardised before release. Impulsive and SS look more step-like with visible gaps between bars, because these two variables take only a small number of distinct values (10 and 11 unique values respectively, as found in Week 2). Impulsive has a slightly longer right tail, while SS is flatter and more spread out than the others.

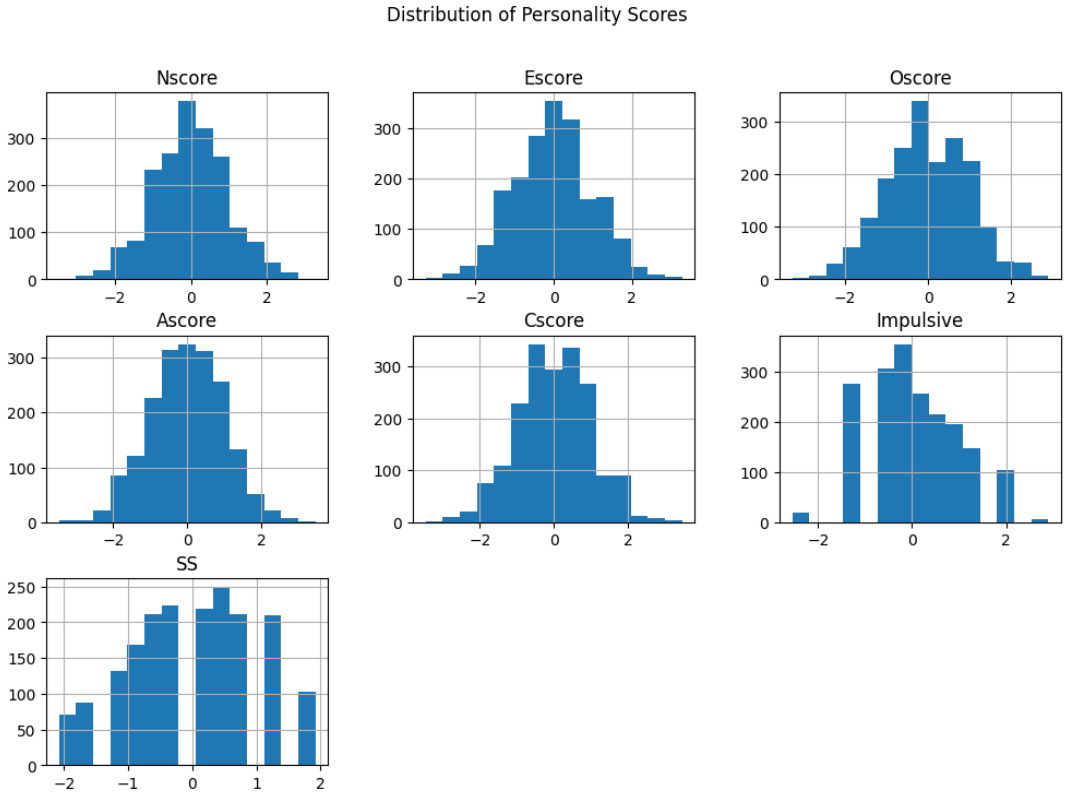


Figure 3: Histograms of the personality scores

5.4 Drug Consumption Distributions

Each drug-consumption variable is recorded as a level from CL0 (never used) to CL6 (used in the last day). Count plots were drawn for nine substances. Their shapes differ clearly, and the main observations are summarised below. Counts are read from the plots and are approximate.

Substance	Most Frequent Level	Approx. Count	Pattern Observed
Alcohol	CL5	760	Concentrated at high levels (CL5, CL6); very few at CL0–CL1
Nicotine	CL6	610	Highest at CL6, with a second large group at CL0
Cannabis	CL6	465	Two-peaked: large groups at both CL6 and CL0
Mushrooms	CL0	985	Mostly never used; CL1–CL3 each have a few hundred
Ecstasy	CL0	1,020	Mostly never used; moderate counts at CL2–CL3
LSD	CL0	1,070	Mostly never used; counts fall steadily across higher levels
Coke	CL0	1,040	Mostly never used; CL2–CL3 form a smaller group
Meth	CL0	1,430	Strongly skewed towards never used
Heroin	CL0	1,600	Most strongly skewed towards never used

Table 4: Summary of drug-consumption distributions

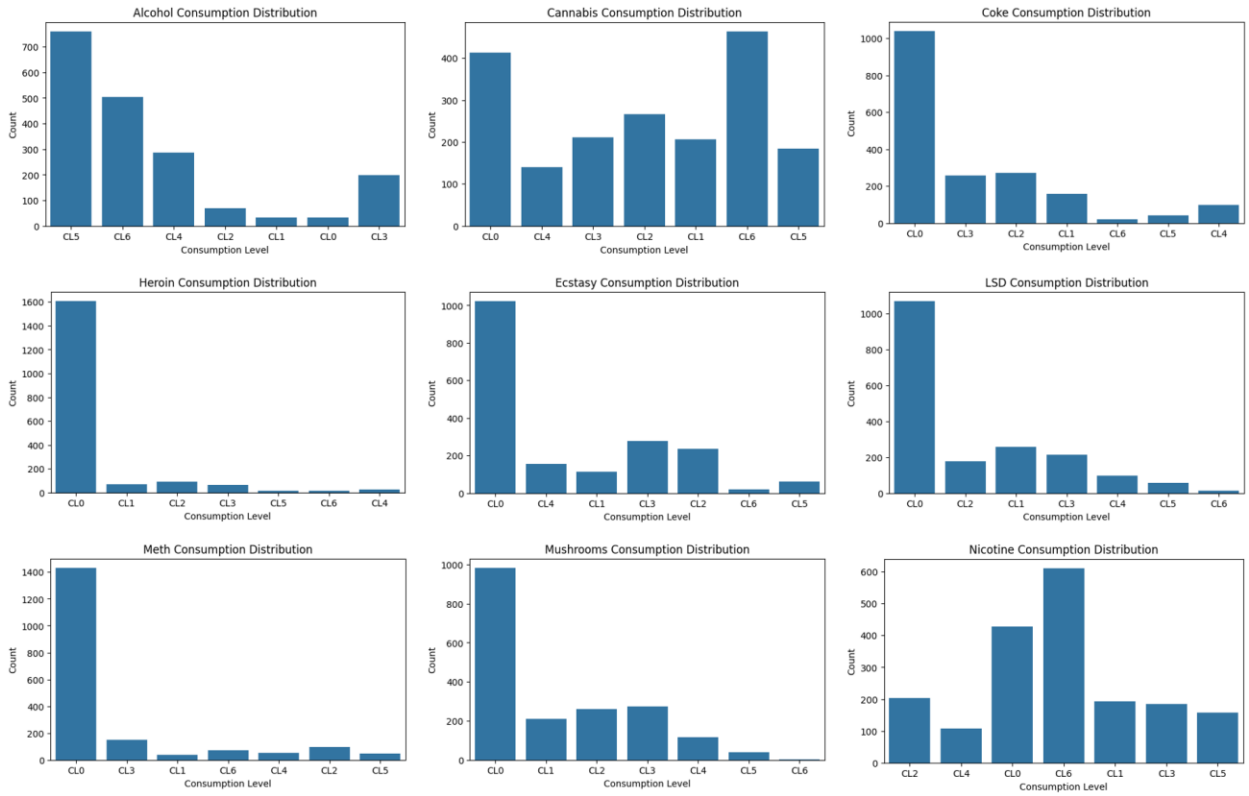


Figure 4: Count plots of drug-consumption levels (Alcohol, Cannabis, Coke, Heroin, Ecstasy, LSD, Meth, Mushrooms, Nicotine)

Alcohol and nicotine are the most widely and most heavily used substances in the data, while heroin and methamphetamine are the least used, with the majority of participants at CL0. Cannabis stands out because its

distribution has two peaks, suggesting that participants tend to be either non-users or very regular users rather than occasional users.

6. Bivariate Analysis

6.1 Age Group vs Gender

A grouped count plot shows how gender is distributed within each age group. In the 18–24 group there are more males (about 365) than females (about 240). From 25–34 to 45–54, females outnumber males, with the gap being widest in the 25–34 group (about 350 females and 213 males). In the 55–64 and 65+ groups the two genders are almost equal. This indicates that the gender balance in the dataset changes with age rather than being uniform.

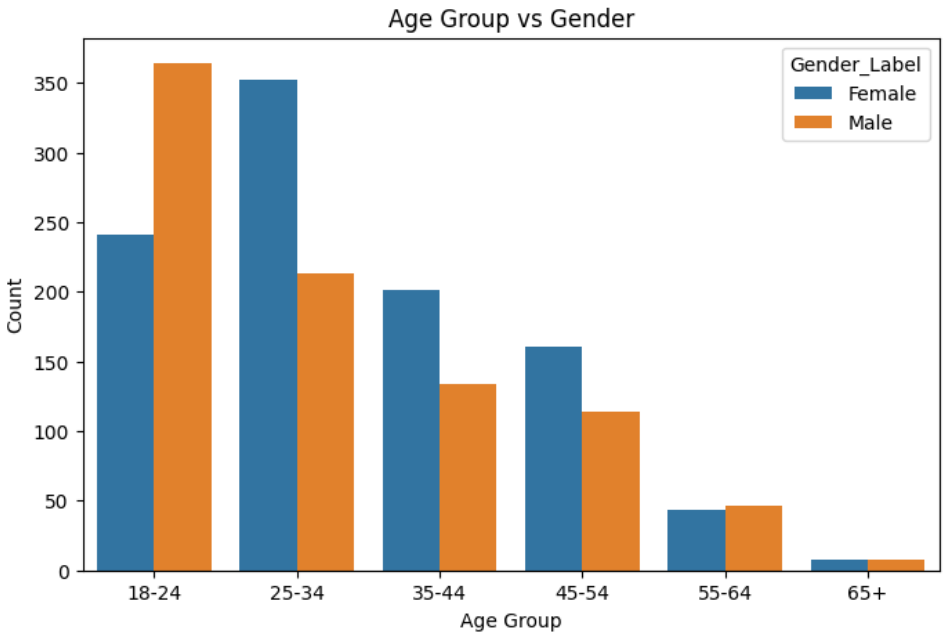


Figure 5: Age Group vs Gender

6.2 Gender vs Alcohol Consumption

Alcohol consumption levels were compared across gender with a grouped count plot. The pattern is very similar for both genders: CL5 is the most common level, followed by CL6 and CL4, while CL0–CL2 are rare. Females have a higher count at CL5 (about 435 compared with about 323 for males) and males have a slightly higher count at CL6 (about 259 compared with about 245), but the overall shape is the same, so gender does not appear to change alcohol use strongly in this dataset.

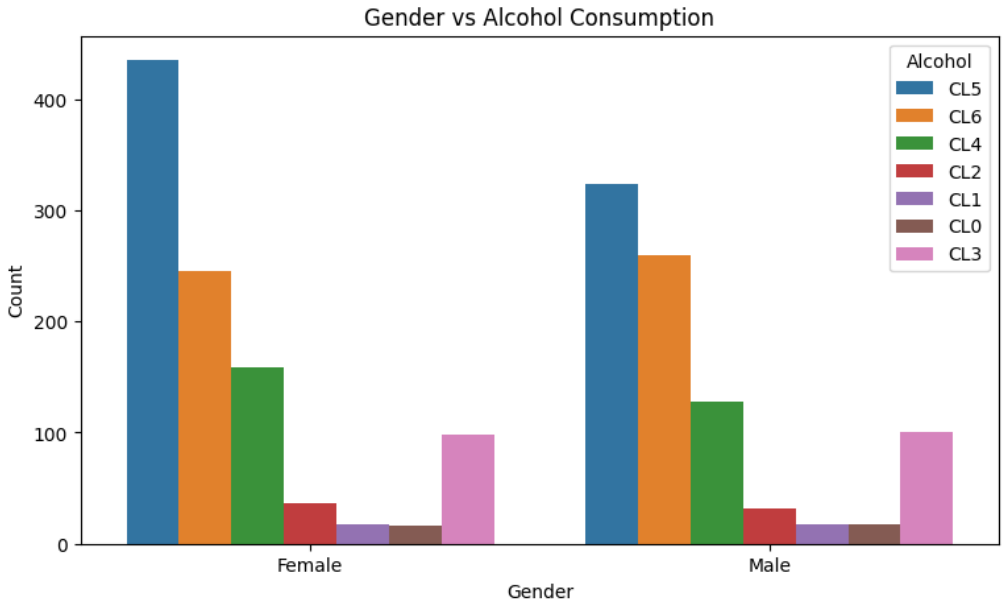


Figure 6: Gender vs Alcohol Consumption

6.3 Age Group vs Alcohol Consumption

The age-group comparison shows that CL5 is the most common alcohol level in most age groups, with CL6 and CL4 next. The counts are highest in the 18–24 and 25–34 groups simply because these groups have the most participants. In the 55–64 group the counts at CL5 and CL6 are almost equal, and the 65+ group is too small to show a reliable pattern. Low consumption levels (CL0–CL2) remain rare in every age group.

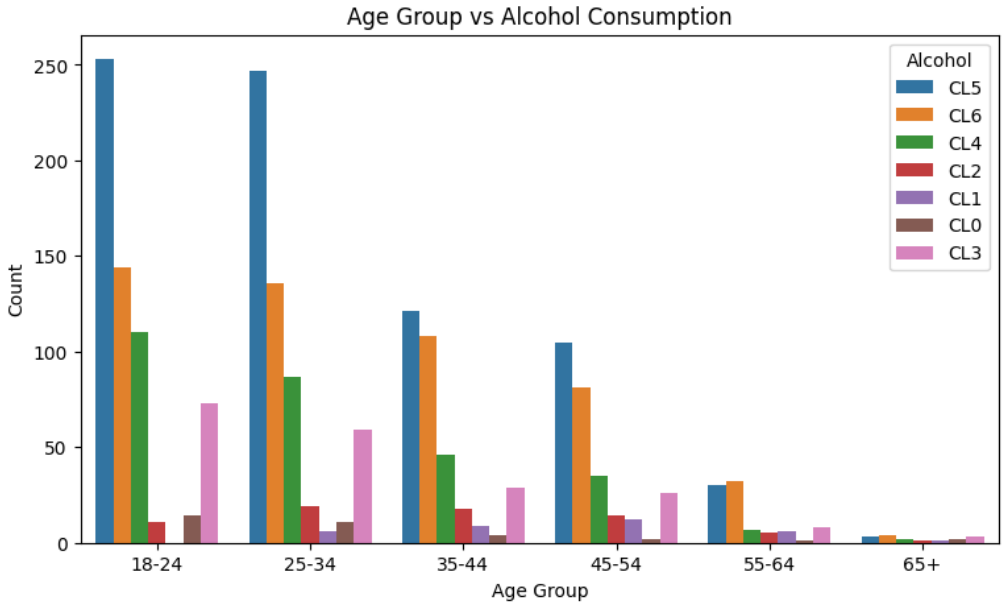


Figure 7: Age Group vs Alcohol Consumption

6.4 Education Level vs Alcohol Consumption

For this plot, the "Median Imputed" education rows were excluded so that only records with an observed education level are compared; the full dataset was not changed. Across all education levels, CL5 and CL6 are the dominant alcohol levels, and the largest bars appear in the education groups that have the most participants. No education level shows a clearly different alcohol pattern, although the smaller groups have too few records for firm conclusions.

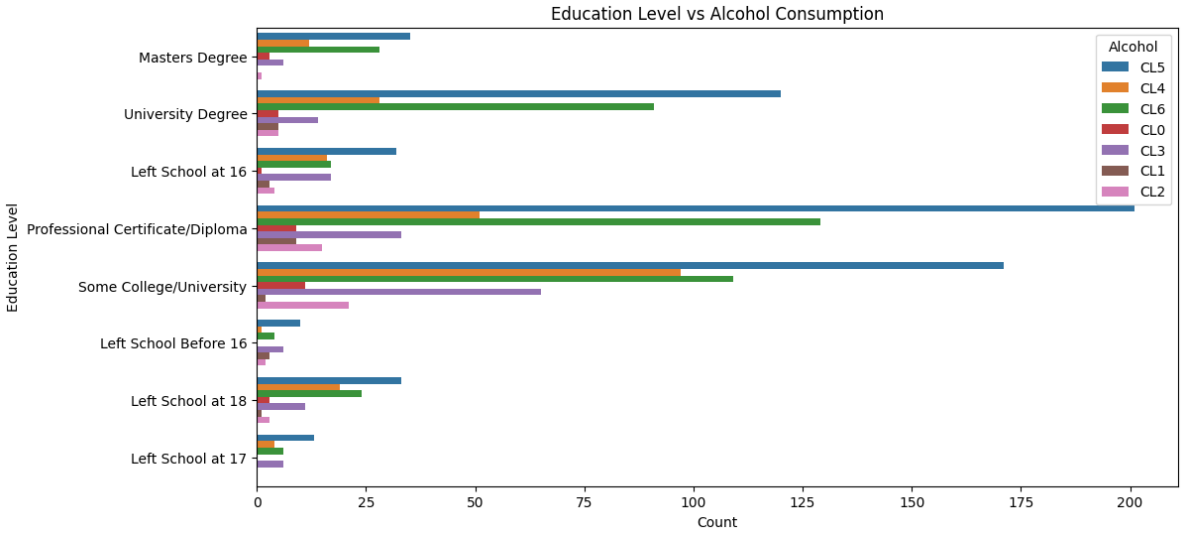


Figure 8: Education Level vs Alcohol Consumption

6.5 Nscore vs Alcohol and Cannabis Consumption

Boxplots compare the numerical Nscore (neuroticism) across consumption levels. For alcohol, the median Nscore stays close to zero for most levels and the boxes overlap heavily, with only small differences: CL2 has a slightly higher median and CL1 a slightly lower one. For cannabis, there is a little more variation, with CL3 showing a higher median Nscore than CL0 and CL1, but the boxes still overlap considerably. Points beyond the whiskers are the potential outliers identified in Week 3. Overall, Nscore alone does not separate the consumption levels strongly.

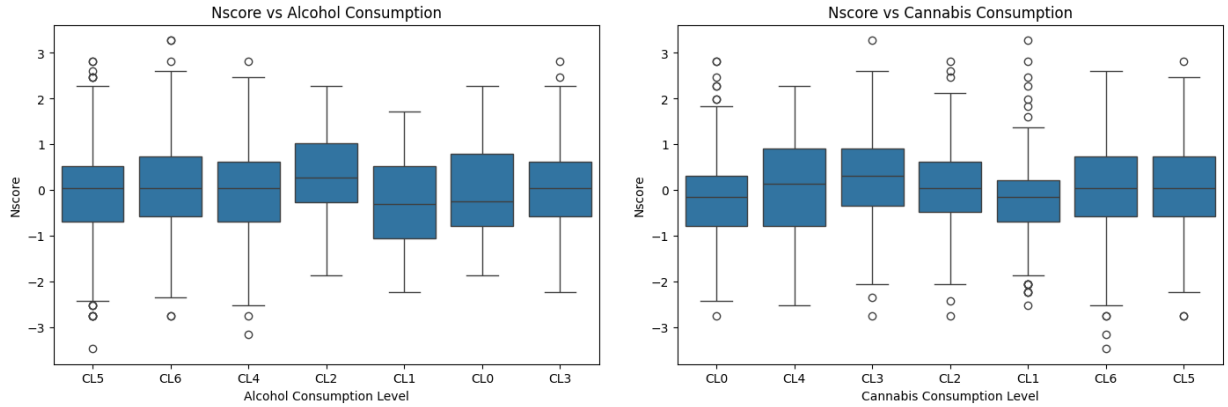


Figure 9: Nscore vs Alcohol Consumption (left) and Nscore vs Cannabis Consumption (right)

6.6 Relationships Between Personality Scores

Scatter plots were used to examine pairs of numerical personality scores. Nscore and Escore show a visible downward trend, meaning that participants with higher neuroticism tend to have lower extraversion (correlation −0.41). Nscore and Oscore show a shapeless cloud with no clear direction (correlation 0.02), so these two traits are essentially unrelated in this dataset.

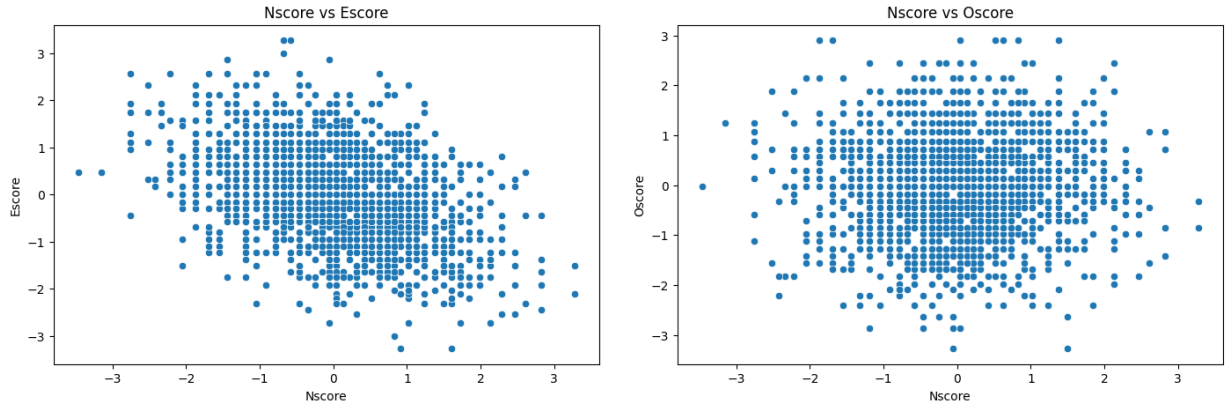


Figure 10: Nscore vs Escore (left) and Nscore vs Oscore (right)

6.7 Correlation Heatmap

A correlation heatmap summarises the strength and direction of the relationship between every pair of personality scores. Values close to +1 indicate a strong positive relationship and values close to −1 a strong negative one. The strongest relationship is between Impulsive and SS (0.62), followed by Oscore and SS (0.42) and Nscore and Escore (−0.41). Most other pairs are weak to moderate, and no pair is strongly correlated enough to suggest that two variables are duplicates of each other.



Figure 11: Correlation heatmap of personality scores

Variable Pair	Correlation	Interpretation
Impulsive – SS	0.62	Moderate to strong positive
Oscore – SS	0.42	Moderate positive
Nscore – Escore	-0.41	Moderate negative
Nscore – Cscore	-0.37	Moderate negative
Cscore – Impulsive	-0.34	Moderate negative
Escore – Cscore	0.31	Moderate positive
Nscore – Oscore	0.02	No relationship

Table 5: Key correlations among personality scores

7. Key Findings

- **Demographics:** females form 53.4% of the data, and the 18–24 age group is the largest (32.1%), which supports the youth focus of the project.
- **Substance use:** alcohol and nicotine are the most commonly used substances, cannabis shows a two-peaked pattern, and heroin and meth are mostly at CL0.
- **Alcohol by group:** the alcohol-consumption pattern (CL5 and CL6 most common) is similar across gender, age group and education level.
- **Personality and consumption:** Nscore differs only slightly across alcohol and cannabis levels, so a single personality score does not explain consumption on its own.
- **Personality relationships:** the strongest correlation is Impulsive–SS (0.62), while Nscore and Escore are moderately negatively related (−0.41).

These findings describe patterns and associations in the data only; they do not show that one variable causes another.

8. Final Data Check

After the label columns were added, the EDA dataset was checked once more. The three new columns (Gender_Label, Age_Group and Education_Level) increase the column count from 32 to 35, while the number of records, the missing values and the duplicate rows remain unchanged.

Check	df_clean	df_eda
Records	1,885	1,885
Columns	32	35
Missing values	0	0
Duplicate rows	0	0

Table 6: Dataset check before and after adding label columns

9. Conclusion

The EDA Deep Dive explored the cleaned dataset using univariate and bivariate analysis. Univariate analysis showed that the data has a slightly higher share of females, is concentrated in the 18–34 age range, has bell-shaped personality scores, and has very different drug-consumption patterns, with alcohol and nicotine used most and heroin and meth used least. Bivariate analysis showed that alcohol consumption is similar across gender, age group and education level, that Nscore differs only slightly across alcohol and cannabis levels, and that some personality traits are related, notably Impulsive with SS and Nscore with Escore. These observations prepare the dataset for the feature engineering stage in the coming weeks.